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METHODS, SYSTEMS, AND DEVICES FOR MONITORING A PORTABLE SPEAKER OF A VEHICLE

Abstract

Methods, systems, and devices for a portable speaker security system. The portable speaker security system may include a portable speaker and a speaker dock. The speaker dock may hold the portable speaker and electrically connect to the portable speaker. The portable speaker security system may further include a head unit electrically connected to the speaker dock. The head unit may indicate a disconnect when the portable speaker is electrically disconnected from the speaker dock. The portable speaker security system may further include a network access device. The network access device may wirelessly communicate with a mobile device. The portable speaker security system may further include an electronic control unit (ECU) electrically connected to the speaker dock, the head unit, and the network access device. The ECU may transmit a message to the mobile device indicating the portable speaker has been disconnected from the speaker dock when the disconnect is indicated.

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Background/Summary

BACKGROUND

1. Field

[0001] The present disclosure relates to methods, systems, and/or devices for monitoring a portable speaker of a vehicle.

2. Description of the Related Art

[0002] Various modern vehicles may include and/or be equipped with a portable speaker (e.g., a portable Bluetooth® speaker). The inclusion of the portable speaker provides convenience to car owners by giving the flexibility of stowing and/or charging the portable speaker within the vehicle but also allowing the car owners to remove the portable speaker from the vehicle (e.g., to use during camping, picnics, hiking, etc.). However, this convenience also presents the risk of the portable speaker being stolen. The risk is especially present with vehicles that may include removable doors and/or a removable roof. Current vehicles do not have systems to monitor a portable speaker to prevent theft of the portable speaker.

[0003] Accordingly, it is desirable to provide methods, systems, and devices for monitoring and/or securing a portable speaker of a vehicle.

SUMMARY

[0004] In general, one aspect of the subject matter described in this disclosure may be embodied in a portable speaker security system for a vehicle. The portable speaker security system may include a portable speaker. The portable speaker security system may further include a speaker dock located on or within the vehicle. The speaker dock may hold the portable speaker and electrically connect to the portable speaker. The portable speaker security system may further include a head unit electrically connected to the speaker dock. The head unit may transmit audio data to the portable speaker through the speaker dock and/or wirelessly transmit the audio data to the portable speaker. The head unit may further indicate a disconnect when the portable speaker is electrically disconnected from the speaker dock. The portable speaker security system may further include a network access device. The network access device may wirelessly communicate with at least one mobile device. The portable speaker security system may further include an electronic control unit (ECU) electrically connected to the speaker dock, the head unit, and the network access device. The ECU may transmit, via the network access device, a message to the at least one mobile device indicating the portable speaker has been disconnected from the speaker dock when the disconnect is indicated.

[0005] In one aspect, the subject matter may be embodied in a portable speaker security system for a vehicle. The portable speaker security system may include a portable speaker. The portable speaker security system may further include a speaker dock located on or within the vehicle. The speaker dock may hold the portable speaker and electrically connect to the portable speaker. The portable speaker security system may further include a network access device. The network access device may wirelessly communicate with at least one mobile device. The portable speaker security system may further include at least one camera. The at least one camera may record an image or a video of a surrounding area of the vehicle or an area within a cabin of the vehicle. The portable speaker security system may further include an electronic control unit (ECU) electrically connected to the speaker dock, the network access device, and the at least one camera. The ECU may indicate a disconnect when the portable speaker is electrically disconnected from the speaker dock. The

ECU may further record, via the at least one camera, the image or the video of the surrounding area of the vehicle or the cabin of the vehicle when the disconnect is indicated. The ECU may further transmit, via the network access device, a message to the at least one mobile device indicating the portable speaker has been disconnected from the speaker dock when the disconnect is indicated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Other systems, methods, features, and advantages of the present disclosure will be apparent to one skilled in the art upon examination of the following figures and detailed description.

Component parts shown in the drawings are not necessarily to scale and may be exaggerated to better illustrate the important features of the present disclosure. In the drawings, like reference numerals designate like parts throughout the different views.

[0007] FIG. 1 is a block diagram of an example portable speaker security system for a vehicle according to an aspect of the disclosure.

[0008] FIG. 2A is a schematic perspective view of a portable speaker of the example portable speaker security system of FIG. 1 in a docked state according to an aspect of the disclosure.

[0009] FIG. 2B is a schematic perspective view of a portable speaker of the example portable speaker security system of FIG. 1 in an undocked state according to an aspect of the disclosure.

[0010] FIG. 3 is a schematic perspective view of a dock of the example portable speaker security system of FIG. 1 according to an aspect of the disclosure.

[0011] FIG. 4 is a schematic side view of a cabin of a vehicle including the example portable speaker security system of FIG. 1 according to an aspect of the disclosure.

[0012] FIG. 5 is a flow diagram of an example process for controlling the portable speaker security system of FIG. 1 according to an aspect of the disclosure.

DETAILED DESCRIPTION

[0013] Disclosed herein are methods, systems, devices, and/or vehicles for implementing a portable speaker security system. Particular embodiments of the subject matter described in this disclosure may be implemented to realize one or more of the following advantages. The portable speaker security system may prevent the unauthorized taking and/or theft of a portable speaker included with and/or located within a vehicle. The portable speaker security system may monitor an electrical connection between the portable speaker and a speaker dock located within the vehicle to determine whether the portable speaker has been removed from the speaker dock. To monitor the electrical connection, a sensor of the portable speaker security system may continuously determine, measure, and/or detect an impedance of the portable speaker. The portable speaker security system may indicate a disconnect in the electrical connection based on the impedance of the portable speaker. When the disconnect is indicated, the portable speaker security system may transmit a message to one or more user devices to notify a user (i.e., an owner and/or authorized user of the vehicle) that the portable speaker has been disconnected from the speaker dock. Hence, the portable speaker security system can advantageously determine if the portable speaker has been misplaced or not on or properly connected to the speaker dock.

[0014] Moreover, the portable speaker security system may activate an alarm of the vehicle when the disconnect is indicated to stop the theft of the portable speaker and/or to notify people within the surrounding area of the vehicle to enhance security of the portable speaker. In examples, the message transmitted to the one or more user devices may provide the user an option to activate and/or deactivate the alarm. In the same and/or additional examples, the message may further include an option to notify the police of the theft of the portable speaker and provide the police with a current location of the vehicle and/or a current location of the portable speaker.

[0015] Moreover, the portable speaker security system may further enhance security of the portable

speaker by including one or more cameras that may record images and/or video of an area within a cabin of the vehicle and/or a surrounding area of the vehicle when the disconnect is indicated. The images and/or the video may track one or more persons running away from the vehicle after the theft has been detected. The images and/or the video may be included in the message that is transmitted to the one or more user devices. The images and/or the video may assist the user and/or the police by showing what happened and/or who removed the portable speaker from the vehicle. [0016] FIG. 1 is a block diagram of an example portable speaker security system **100** (also can be referred to as a portable device security system **100**). The portable speaker security system **100** or a portion thereof may be retrofitted, coupled to, include, or be included within a vehicle **102** or separate from the vehicle **102**. The vehicle **102** may be a conveyance capable of transporting a person, an object, or a permanently or temporarily affixed apparatus. The vehicle **102** may be a self-propelled wheeled conveyance, such as a car, a sports utility vehicle, a truck, a bus, a van or other motor, battery or fuel cell driven vehicle. For example, the vehicle **102** may be an electric vehicle, a hybrid vehicle, a hydrogen fuel cell vehicle, a plug-in hybrid vehicle or any other type of vehicle that has a fuel cell stack, a motor, an engine, and/or a generator. Other examples of vehicles include bicycles, trains, planes, or boats, and any other form of conveyance that is capable of transportation. The vehicle **102** may be semi-autonomous or autonomous. That is, the vehicle **102** may be self-maneuvering and navigate without human input. An autonomous vehicle may have and use one or more sensors and/or a navigation unit to drive autonomously.

[0017] The portable speaker security system **100** and/or the vehicle **102** may include a motor and/or generator **132** and a battery **134**. The motor and/or generator **132** may be located within an engine bay of the vehicle **102**. The motor and/or generator **132** may be an internal combustion engine (ICE). In this regard, the motor and/or generator **132** may combust an air and fuel mixture to provide power to the vehicle **102** and/or components of the vehicle **102** and/or the portable speaker security system **100**. Accordingly, the motor and/or generator **132** can cause the vehicle **102** to accelerate, decelerate, or maintain a desired velocity. It should be understood that the motor and/or generator **132** may include combinations of an ICE and an electric motor, such as for hybrid vehicle applications for example. In examples, the motor and/or generator **132** may be an electric motor. In this regard, the motor and/or generator **132** may be an electric motor and an electric generator that converts electrical energy into mechanical power, such as torque, and converts mechanical power into electrical energy. The motor and/or generator **132** may be electrically connected to the battery **134**. The motor and/or generator **132** may convert energy from the battery **134** into mechanical power, and may provide energy back to the battery **134**, for example, via regenerative braking. The battery **134** may be electrically connected to the motor and/or generator **132** and may provide electrical energy to and/or receive electrical energy from the motor and/or generator **132**. The battery **134** may provide electrical energy to the portable speaker security system **100**.

[0018] The portable speaker security system **100** may further include an alarm device **136**. The alarm device **136** may be located within the vehicle **102**. The alarm device **136** may be an integrated alarm system of the vehicle **102** and may be and/or include a horn of the vehicle **102** and/or one or more lights of the vehicle **102**. When activated, the alarm device **136** may continuously and/or repeatedly generate sound (e.g., by sounding the horn of the vehicle **102**) and/or may flash the one or more lights of the vehicle **102**.

[0019] FIGS. 2A and 2B are schematic perspective views of a portion of the example portable speaker security system **100** of FIG. 1 within a cabin **202** of the vehicle **102**. With combined reference to FIGS. 1, 2A, and 2B, the portable speaker security system **100** may further include a portable speaker **104** and a dock **105** (e.g., a speaker dock).

[0020] The portable speaker **104** may be removably coupled to the dock **105**. The portable speaker **104** may be located within the cabin **202** and/or a dashboard **204** of the vehicle **102** and may be moved to outside the cabin **202** by a user. For example, the portable speaker **104** can be positioned

within the dashboard **204** so that the speaker **104** is flush with a top surface of the dashboard **204** when inserted and connected. The portable speaker **104** may be a wireless speaker configured to receive audio data wirelessly (e.g., via Bluetooth®, Wi-Fi, and/or other wireless technologies) from one or more user devices **142** (e.g., a mobile phone, a laptop, a tablet, etc.) and/or over a wired connection (e.g., via an audio data port) from the one or more user devices **142** and/or the vehicle **102**. The portable speaker **104** may include a rechargeable battery **201** for powering the portable speaker **104**.

[0021] The portable speaker **104** may have a docked state (or position) **208**, as shown by FIG. 2A, and an undocked state (or position) **210**, as shown by FIG. 2B. The portable speaker **104** may receive electrical energy for charging the rechargeable battery **201** of the portable speaker **104** and/or audio data from the dock **105** when the portable speaker **104** is in the docked state **208**. In examples, the portable speaker **104** may include one or more terminals (or electrical contacts) **216** for receiving the electrical energy and/or the audio data from the dock **105**. The one or more terminals **216** of the portable speaker **104** may be and/or include one or more power connectors and/or one or more audio (or data) connectors. The portable speaker **104** may be placed in the dock **105** and may be held within the dock **105**. In examples, the portable speaker **104** may include one or more receiving portions **212** configured to receive and/or interface with one or more tabs (or retention tabs) **214** of the dock **105** to hold the portable speaker **104** within the dock **105**. The one or more receiving portions **212** may include a magnet, a fastener, or a clip to hold the portable speaker **104** within the dock **105**.

[0022] FIG. 3 is a schematic perspective view of the dock **105** of the portable speaker security system **100**. With combined reference to FIGS. 2A, 2B, and 3, the dock **105** may be coupled to the dashboard **204** and/or located within the cabin **202** of the vehicle **102**. The dock **105** may receive, hold, secure, and/or release the portable speaker **104** to removably couple the portable speaker **104** to the dashboard **204** and/or the cabin **202** of the vehicle **102**. The dock **105** may include one or more terminals (or electrical contacts) **304** for providing the electrical energy and/or the audio data to the portable speaker **104**. The one or more terminals **304** of the dock **105** may be and/or include one or more power connectors and/or one or more audio (or data) connectors for providing the electrical energy and/or the audio data to the portable speaker **104**. The one or more terminals **304** of the dock **105** may electrically connect with the one or more terminals **216** of the portable speaker **104** when the portable speaker **104** is in the docked state **208**.

[0023] To release the portable speaker **104** from the dock **105**, a user may press on a release button **206** of the dock **105**, as shown by FIG. 2B. In examples, when the release button **206** is pressed, the one or more tabs **214** of the dock **105** may retract from the one or more receiving portions **212** of the portable speaker **104** thereby releasing the portable speaker **104** from the dock **105**. In examples, the dock **105** may include ejectors (or spring loaded ejectors) **306** configured to push the portable speaker **104** out of the dock **105**. The spring loaded ejectors **306** may apply a constant force against the portable speaker **104** when the portable speaker **104** is in the docked state **208** and/or may be activated to push the portable speaker **104** out of the dock **105** by pressing the release button **206**.

[0024] FIG. 4 is a schematic side view of the cabin **202** of the vehicle **102** including the example portable speaker security system **100** of FIG. 1. With combined reference to FIGS. 1 and 4, the portable speaker security system **100** may further include a head unit **120**. The head unit **120** may be located within and/or coupled to the dashboard **204**. The head unit **120** may be electrically connected to the dock **105** and one or more speakers **402** of the vehicle **102**. The head unit **120** may control the one or more speakers **402** and/or the portable speaker **104**. For example, the head unit **120** may provide audio data and/or electrical energy to the one or more speakers **402** and/or may provide the audio data and/or the electrical energy to the portable speaker **104** via the dock **105** when the portable speaker **104** is in the docked state **208**. In examples, when the portable speaker **104** is in the docked state **208**, the head unit **120** may utilize the portable speaker **104** as a central

channel such that the portable speaker **104** is included in the one or more speakers **402** of the vehicle **102**. In examples, when the portable speaker **104** is disconnected from the dock **105**, the head unit **120** may wireless provide the audio data to the portable speaker **104**. The head unit **120** may further include one or more digital to analog converters (DACs) and/or one or more amplifiers to process and/or control the audio data. In examples, the head unit **120** may further include a microcontroller (or audio microcontroller) **122** for controlling the one or more speakers **402** and/or the portable speaker **104**.

[0025] The portable speaker security system **100** may further include an impedance (or resistance) sensor (or circuit) **109**. The impedance sensor **109** may be electrically connected to the head unit **120**, the dock **105**, and/or the portable speaker **104**. In examples, the impedance sensor **109** may be included within the dashboard **204**, the dock **105**, and/or the head unit **120**. The impedance sensor **109** may measure, detect, and/or determine an impedance of the portable speaker **104** and/or a circuit that is completed when the portable speaker **104** is electrically connected to the speaker dock **105**. For example, when the portable speaker **104** is in the docked state **208**—and therefore electrically connected to the dock **105**—the impedance sensor **109** may detect an impedance of 1-10 ohms (in examples, the detected impedance may be between 1-100 ohms). When the portable speaker **104** is removed from the dock **105** such that the portable speaker **104** is not electrically connected to the dock **105** (i.e., the portable speaker **104** is in the undocked state **210**), the impedance sensor **109** may detect an infinite and/or near infinite impedance because of an open circuit created by removing the portable speaker **104** from the dock **105** (e.g., an open circuit at and/or between the one or more terminals **304** of the dock **105**).

[0026] The portable speaker security system **100** may further include one or more cameras **138**. The one or more cameras **138** may be located within the cabin **202** and/or on an exterior of the vehicle **102**. In examples, the one or more cameras **138** may be coupled to one or more side view mirrors **404**, a rear view mirror **406**, a ceiling of the cabin **202**, and/or the dashboard **204** of the vehicle **102**. The one or more cameras **138** may be a digital camera, an infrared thermal camera, and/or a night vision camera (e.g., utilizing active illumination and/or image intensification). The one or more cameras **138** may provide, capture, and/or record images and/or video of an area within the cabin **202** and/or a surrounding area of the vehicle **102**.

[0027] The portable speaker security system **100** may further include a network access device **114**. In examples, the network access device **114** may be a telematic control unit (TCU). The network access device **114** may include a communication port or channel, such as one or more of a Wi-Fi unit, a Bluetooth® unit, a Radio Frequency Identification (RFID) tag or reader, a DSRC unit, and/or a cellular network unit for accessing a network **140** (e.g., CDMA, GSM, 3G, 4G, 5G, etc.). The network access device **114** may transmit data to and receive data from devices and systems not directly connected to the vehicle **102**. For example, an electronic control unit (ECU) **106** of the vehicle **102** may communicate with one or more user devices **142** (e.g., a key fob, a mobile device, phone, tablet, laptop, etc.) through the network access device **114**. In examples, a user (e.g., the owner of the vehicle **102**) may use the one or more user devices **142** to communicate with the ECU **106** to start the vehicle **102** and/or to activate/deactivate the portable speaker security system **100**.

[0028] The portable speaker security system **100** may further include a navigation unit **118**. The navigation unit **118** may provide vehicle information (or data) and/or navigational map information to the ECU **106**. The navigation unit **118** may have and/or be connected to a Global Positioning System (GPS) device. The vehicle information may include a current position and/or location of the vehicle **102**, a current time at the current position, a direction of travel, and/or a current speed of the vehicle **102**.

[0029] The portable speaker security system **100** may further include one or more processors and/or microcontrollers, such as the electronic control unit (ECU) **106** and the microcontroller **122**. The ECU **106** may be implemented as a single ECU or in multiple ECUs, with each ECU of the multiple ECUs controlling one or more components of the vehicle **102** and/or the portable speaker

security system **100**. The ECU **106** may be electrically connected to some or all of the components of the vehicle **102** and/or the portable speaker security system **100** via a local area network (LAN) such as, but not limited to, a vehicle bus utilizing one or more protocols (e.g., controller area network (CAN), local interconnect network (LIN), etc.). The ECU **106** may be electrically connected to the alarm device **136**, the portable speaker **104**, the dock **105**, the head unit **120**, the impedance sensor **109**, the one or more cameras **138**, the network access device **114**, the navigation unit **118**, a memory **108**, and/or a user interface **110**. The ECU **106** may include one or more processors (or controllers) specifically designed for controlling operations of the vehicle **102** such as accelerating, braking, etc. In examples, the ECU **106** may be and/or include the microcontroller **122** within the head unit **120**.

[0030] The portable speaker security system **100** may monitor an electrical connection between the portable speaker **104** and the dock **105** to determine whether the portable speaker **104** has been removed from the dock **105**. In examples, the electrical connection may be formed by the one or more terminals **304** of the dock **105** electrically connecting to the one or more terminals **216** of the portable speaker **104**. To monitor the electrical connection, the impedance sensor **109** may continuously determine, measure, and/or detect an impedance of the portable speaker **104** and/or the impedance between the one or more terminals **304** of the dock **105** and/or between the one or more terminals **304** of the dock **105** and the one or more terminals **216** of the portable speaker **104**. The impedance sensor **109** may output an impedance reading (or impedance data) corresponding to the impedance of the portable speaker **104** and/or the impedance between the one or more terminals **304** of the dock **105** and/or between the one or more terminals **304** and the one or more terminals **216** of the portable speaker **104**) to the ECU **106** and/or the head unit **120**.

[0031] The ECU **106** and/or the head unit **120** (e.g., the microcontroller **122** of the head unit **120**) may receive the impedance reading and may indicate a disconnect in the electrical connection when the impedance reading is infinite and/or greater than an impedance threshold (i.e., when the portable speaker **104** is electrically disconnected from the dock **105**). In response to the disconnect being indicated, the ECU **106** may transmit, via the network access device **114**, a message (or information) to one or more user devices **142** (e.g., a portable electronic device or a phone of an owner of the vehicle **102**) indicating the portable speaker **104** has been disconnected from the dock **105**.

[0032] In examples, when the disconnect is indicated, the ECU **106** may record, via the one or more cameras **138**, one or more images and/or one or more videos of the area within the cabin **202** and/or the surrounding area of the vehicle **102** to capture an image of the person who removed the portable speaker **104** from the dock **105**. In examples, the message may include the one or more images and/or the one or more videos to allow a user of the one or more user devices **142** to see who removed the portable speaker **104** from the dock **105**.

[0033] The portable speaker security system **100** may further include the memory **108**. The memory **108** may be electrically connected to the ECU **106**. In examples, the memory **108** may be communicatively coupled (e.g., via the network **140**) to the ECU **106** such that the memory **108** is remote from the ECU **106** and/or the vehicle **102**. In other examples, the memory **108** may be electrically connected to the ECU **106** and a remote memory may be communicatively coupled to the ECU **106**, with the remote memory having similar, additional, and/or different functions as the memory **108** (e.g., greater storage capacity, enabling over-the-air updates, etc.). The memory **108** may store instructions to execute on the ECU **106** and may include one or more of a random access memory (RAM) or other volatile or non-volatile memory. The memory **108** may be a non-transitory memory or a data storage device, such as a hard disk drive, a solid-state disk drive, a hybrid disk drive, or other appropriate data storage, and may further store machine-readable instructions, which may be loaded and executed by the ECU **106**. The memory **108** may store vehicle parameters (e.g., vehicle weight, transmission gear information, etc.). In examples, the memory **108** may further store the images and/or the video captured by the one or more cameras

138. In the same and/or additional examples, the memory **108** may store the impedance threshold. The impedance threshold may be between 1-10 ohms and/or may be any impedance corresponding to the portable speaker **104** being in the undocked state **210**. In one aspect, the impedance threshold may be about or a little over 50 ohms (e.g., 55 ohms) to indicate that the portable speaker **104** is connected to the dock **105**.

[0034] The portable speaker security system **100** may further include the user interface **110**. The user interface **110** may provide an interface to a user of the vehicle **102** to interact with and/or receive output from the ECU **106**. The user interface **110** may have a user interface element, such as a screen and/or a touchscreen with a button, a switch, a microphone, a speaker, a gesture monitoring sensor, a knob, a graphical user interface (GUI), and/or other input/output devices electrically connected to the ECU **106** to provide input and/or output of information (or data) to and/or from the ECU **106**.

[0035] In examples, the user interface **110** and/or the one or more user devices **142** may enable a user to change various settings of the portable speaker security system **100**. For example, the user may set the portable speaker security system **100** to activate, via the ECU **106**, the alarm device **136** when the disconnect is indicated. In another example, the user may set when the portable speaker security system **100** is active (i.e., monitoring the electrical connection between the dock **105** and the portable speaker **104**). The user may set the portable speaker security system **100** to be active, for example, when the vehicle **102** is locked and/or when a key (e.g., a key card, a key fob, a mobile phone acting as a key, etc.) is not detected by the ECU **106** and/or within a predetermined range (e.g., 5 feet, 10 feet, 20 feet, etc.) of the vehicle **102**. The user interface **110** and/or the one or more user devices **142** may further enable the user to enter a passcode to deactivate the portable speaker security system **100** and/or prevent the message from being transmitted to the one or more user devices **142** such that the user may remove the portable speaker **104** from the dock **105** without activating the portable speaker security system **100** and/or the alarm device **136**. In examples, the user interface **110** and/or the one or more user devices **142** may further prompt the user to enter the passcode to deactivate the alarm device **136** after the alarm device **136** has been activated.

[0036] FIG. 5 is a flow diagram of an example process **500** for monitoring and/or securing a portable speaker of a vehicle. One or more computers or one or more data processing apparatuses, for example, the ECU **106** and/or the microcontroller **122** of the portable speaker security system **100** of FIG. 1, appropriately programmed, may implement the process **500**. For ease of description, the process **500** is described below with reference to FIGS. 1, 2A, 2B, 3, and 4. The process **500** of the present disclosure, however, is not limited to use of the exemplary portable speaker security systems of FIGS. 1, 2A, 2B, 3, and 4.

[0037] The portable speaker security system **100** may detect and/or monitor, via the impedance sensor **109**, an impedance reading of the portable speaker **104** electrically connected to the dock **105** (**502**). The impedance sensor **109** may output the impedance reading (or impedance data) corresponding to the impedance of the portable speaker **104** and/or the impedance between the one or more terminals **304** to the ECU **106** and/or the head unit **120**.

[0038] The portable speaker security system **100** may indicate (e.g., via the ECU **106** and/or the head unit **120**) a disconnect between the portable speaker **104** and the dock **105** when the impedance reading is greater than a threshold (or impedance threshold) (**504**). The threshold may be predetermined and may be stored in the memory **108**. The threshold may be 1-500 ohms (or another impedance that corresponds to the portable speaker **104** being in the docked state **208** and electrically connected to the dock **105**).

[0039] In response to the disconnect being indicated, the portable speaker security system **100** may activate the alarm device **136** of the vehicle **102**.

[0040] The portable speaker security system **100** may transmit, via the network access device **114** a message (or information) to one or more user devices **142** (e.g., a phone of an owner of the vehicle

102) indicating the portable speaker **104** has been disconnected from the dock **105** (**508**). In examples, the message may include a user selectable option indicating the disconnect was unauthorized, with the alarm device **136** being activated upon, and not before, the portable speaker security system **100** receiving, via the network access device **114**, data from the one or more user devices **142** indicating the disconnect was unauthorized (i.e., the user selectable option was selected by the user). In the same or additional examples, after receiving the data from the one or more user devices **142** indicating the disconnect was unauthorized, the portable speaker security system **100** may transmit another message to police indicating the portable speaker **104** has been stolen and indicating, via the navigation unit **118**, a current location of the vehicle **102**. In the same or additional examples, the message may include another user selectable option indicating the disconnect was authorized. When the another user selectable option is selected by the user, the portable speaker security system **100** may deactivate the alarm device **136** if the alarm device **136** has been activated and/or may prevent the alarm device **136** from being activated.

[0041] In examples, the portable speaker security system **100** may stop and/or prevent the message from being transmitted to the one or more user devices **142** when a key of the vehicle **102** is detected by the vehicle **102** and/or within a predetermined distance from the vehicle **102**.

[0042] In examples, when the disconnect is indicated, the portable speaker security system **100** may record, via the one or more cameras **138**, one or more images and/or one or more videos of the area within the cabin **202** of the vehicle **102** and/or the surrounding area of the vehicle **102** (**510**).

[0043] In examples, the portable speaker security system **100** may further transmit, via the network access device **114**, the one or more images and/or the one or more videos to the one or more user devices **142** (**512**).

[0044] Exemplary embodiments of the invention have been disclosed in an illustrative style. Accordingly, the terminology employed throughout should be read in a non-limiting manner. Although minor modifications to the teachings herein will occur to those well versed in the art, it shall be understood that what is intended to be circumscribed within the scope of the patent warranted hereon are all such embodiments that reasonably fall within the scope of the advancement to the art hereby contributed, and that that scope shall not be restricted, except in light of the appended claims and their equivalents.

Claims

1. A portable speaker security system for a vehicle, comprising: a portable speaker; a speaker dock located on or within the vehicle and configured to hold the portable speaker and electrically connect to the portable speaker; a head unit electrically connected to the speaker dock and configured to: transmit audio data to the portable speaker through the speaker dock and/or wirelessly transmit the audio data to the portable speaker, and indicate a disconnect when the portable speaker is electrically disconnected from the speaker dock; a network access device configured to wirelessly communicate with at least one mobile device; and an electronic control unit (ECU) electrically connected to the speaker dock, the head unit, and the network access device and configured to: transmit, via the network access device, a message to the at least one mobile device indicating the portable speaker has been disconnected from the speaker dock when the disconnect is indicated.

2. The portable speaker security system of claim 1, further comprising at least one camera electrically connected to the ECU, wherein the ECU is further configured to: record, via the at least one camera, an image or a video of a surrounding area of the vehicle and/or an area within a cabin of the vehicle when the disconnect is indicated.

3. The portable speaker security system of claim 2, wherein the ECU is further configured to transmit, via the network access device, the image or the video to the at least one mobile device.

4. The portable speaker security system of claim 1, further comprising an alarm device electrically

connected to the ECU and configured to generate sound when the alarm device is activated, wherein the ECU is further configured to: activate the alarm device when the disconnect is indicated.

5. The portable speaker security system of claim 1, further comprising an alarm device electrically connected to the ECU and configured to generate sound when the alarm device is activated, wherein: the message includes a user selectable option indicating the disconnect was unauthorized; and the ECU is further configured to: receive, via the network access device, data indicating the user selectable option was selected by a user, and activate the alarm device in response to receiving the data.

6. The portable speaker security system of claim 1, wherein: the message includes a user selectable option indicating the disconnect was unauthorized; and the ECU is further configured to: receive, via the network access device, data indicating the user selectable option was selected, and transmit another message to police indicating the portable speaker has been stolen in response to receiving the data.

7. The portable speaker system of claim 5, further comprising a navigation unit electrically connected to the ECU and configured to determine a current location of the vehicle, wherein the another message includes the current location of the vehicle.

8. The portable speaker security system of claim 1, further comprising an impedance sensor configured to detect an impedance reading of the portable speaker and/or a circuit that is completed when the portable speaker is electrically connected to the speaker dock, wherein the head unit is further configured to determine the portable speaker is electrically disconnected from the speaker dock when the impedance reading is greater than a threshold impedance.

9. The portable speaker security system of claim 1, wherein: the ECU is further configured to stop transmitting the message to the at least one mobile device when a vehicle key is within a predetermined distance from the vehicle.

10. The portable speaker security system of claim 1, further comprising a user interface electrically connected to the ECU, wherein the ECU is further configured to: display, via the user interface, a prompt to enter a passcode when the disconnect is indicated, and stop transmitting the message to the at least one mobile device when the passcode is entered.

11. A portable speaker security system for a vehicle, comprising: a portable speaker; a speaker dock located on or within the vehicle and configured to hold the portable speaker and electrically connect to the portable speaker; a network access device configured to wirelessly communicate with at least one mobile device; at least one camera configured to record an image or a video of a surrounding area of the vehicle and/or an area within a cabin of the vehicle; an electronic control unit (ECU) electrically connected to the speaker dock, the network access device, and the at least one camera and configured to: indicate a disconnect when the portable speaker is electrically disconnected from the speaker dock, record, via the at least one camera, the image or the video of the surrounding area of the vehicle and/or the area within the cabin of the vehicle when the disconnect is indicated, and transmit, via the network access device, a message to the at least one mobile device indicating the portable speaker has been disconnected from the speaker dock when the disconnect is indicated.

12. The portable speaker security system of claim 11, wherein the ECU is further configured to transmit, via the network access device, the image or the video to the at least one mobile device.

13. The portable speaker security system of claim 11, further comprising an alarm device electrically connected to the ECU and configured to generate sound when the alarm device is activated, wherein the ECU is further configured to: activate the alarm device when the disconnect is indicated.

14. The portable speaker security system of claim 11, further comprising an alarm device electrically connected to the ECU and configured to generate sound when the alarm device is activated, wherein: the message includes a user selectable option indicating the disconnect was

unauthorized; and the ECU is further configured to: receive, via the network access device, data indicating the user selectable option was selected by a user, and activate the alarm in response to receiving the data.

15. The portable speaker security system of claim 11, wherein: the message includes a user selectable option indicating the disconnect was unauthorized; and the ECU is further configured to: receive, via the network access device, data indicating the user selectable option was selected, and transmit another message to police indicating the portable speaker has been stolen in response to receiving the data.

16. The portable speaker system of claim 15, further comprising a navigation unit electrically connected to the ECU and configured to determine a current location of the vehicle, wherein the another message includes the current location of the vehicle.

17. The portable speaker security system of claim 11, wherein: the ECU is further configured to stop transmitting the message to the at least one mobile device when a vehicle key is within a predetermined distance from the vehicle.

18. The portable speaker security system of claim 11, further comprising a user interface electrically connected to the ECU, wherein the ECU is further configured to: display, via the user interface, a prompt to enter a passcode when the disconnect is indicated, and stop transmitting the message to the at least one mobile device when the passcode is entered.

19. The portable speaker security system of claim 11, further comprising: a head unit electrically connected to the speaker dock and the ECU and configured to transmit audio data to the portable speaker through the speaker dock and/or wirelessly transmit the audio data to the portable speaker.

20. The portable speaker security system of claim 19, wherein: the speaker dock includes at least two terminals configured to electrically connect to the portable speaker; the head unit includes an impedance sensor configured to detect an impedance reading of the portable speaker and/or a circuit that is completed when the portable speaker is electrically connected to the dock; and the ECU is further configured to determine the portable speaker is electrically disconnected from the speaker dock when the impedance reading is greater than a threshold impedance.
